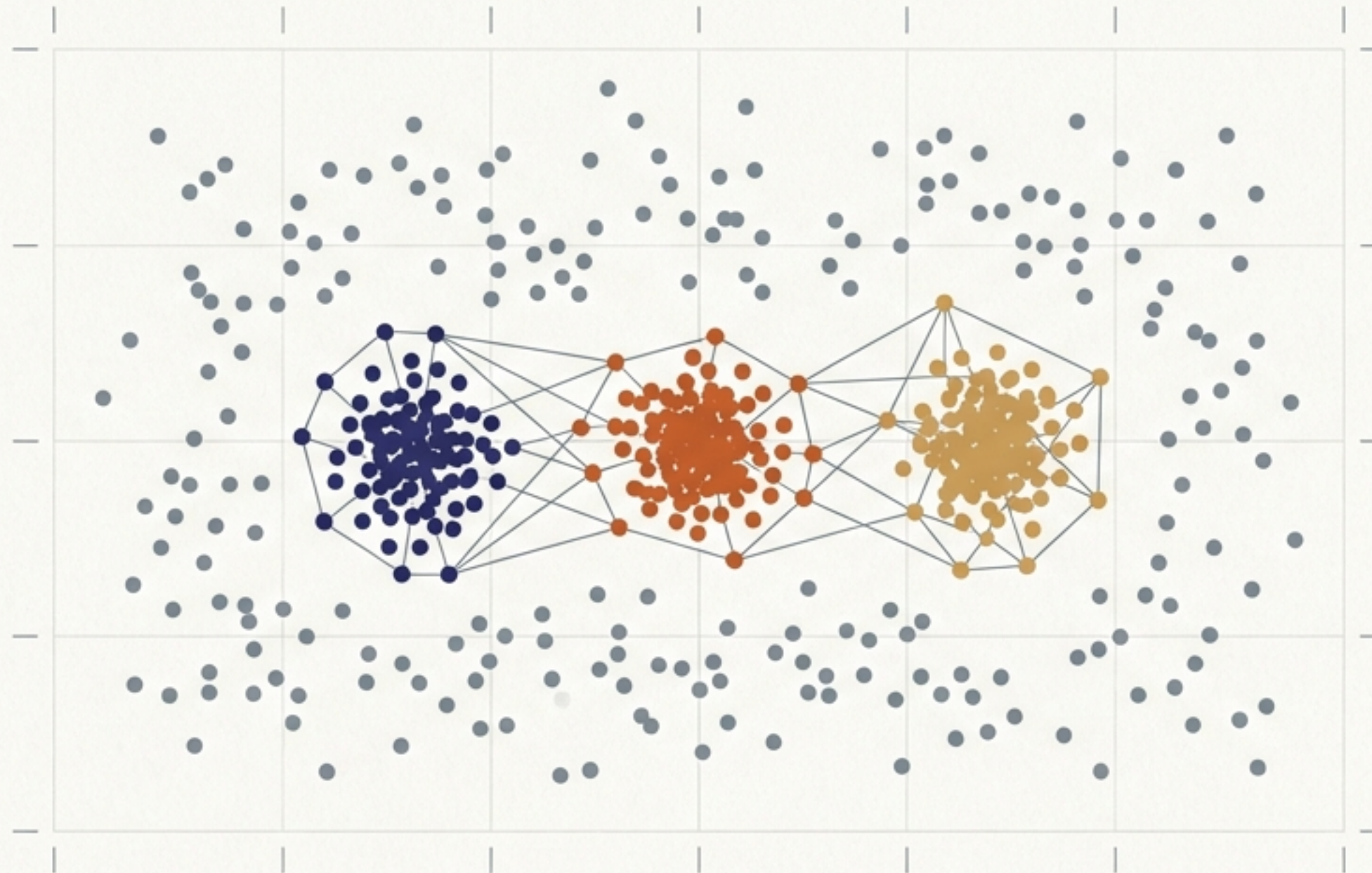


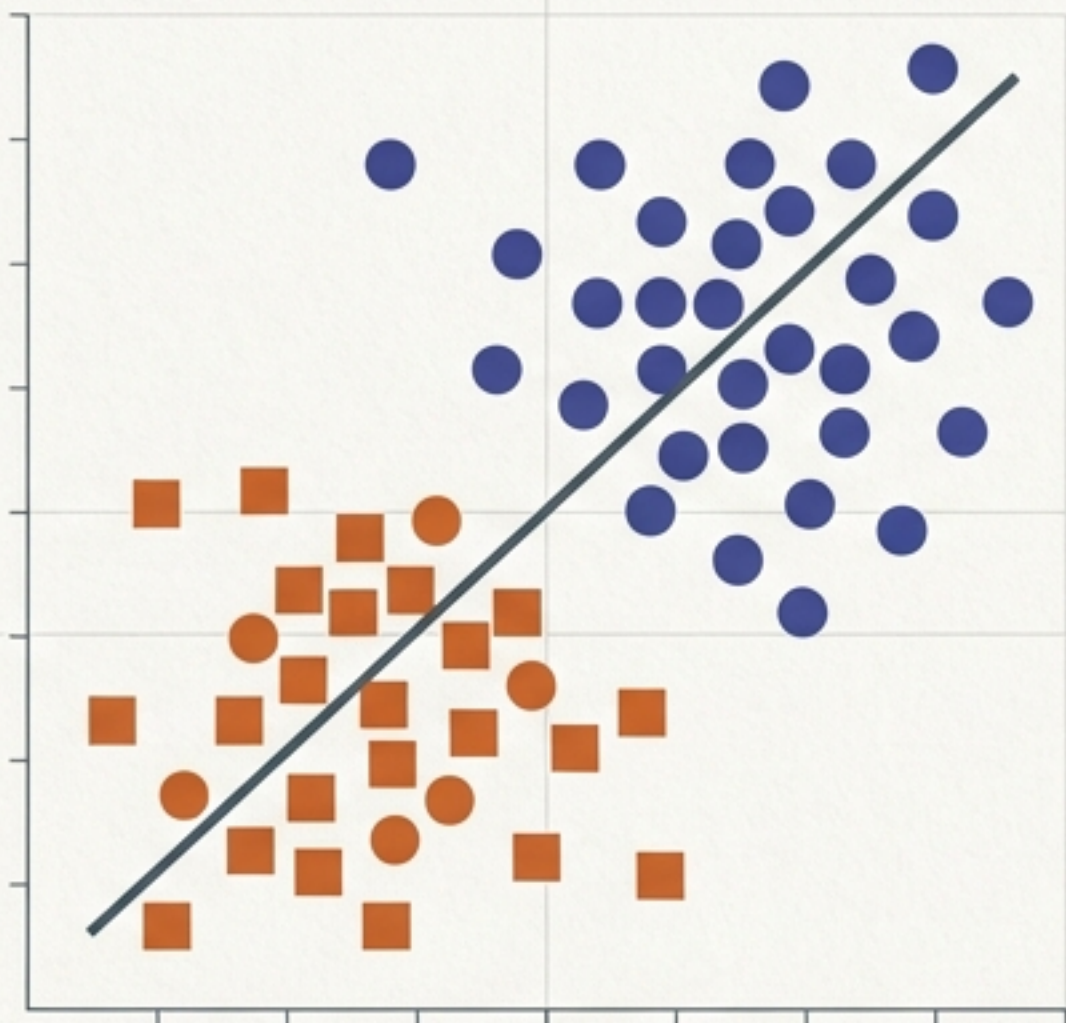


# Mapping the Unknown: Machine Learning Clustering

A Deep Dive into K-Means & Hierarchical Algorithms for Unsupervised Learning



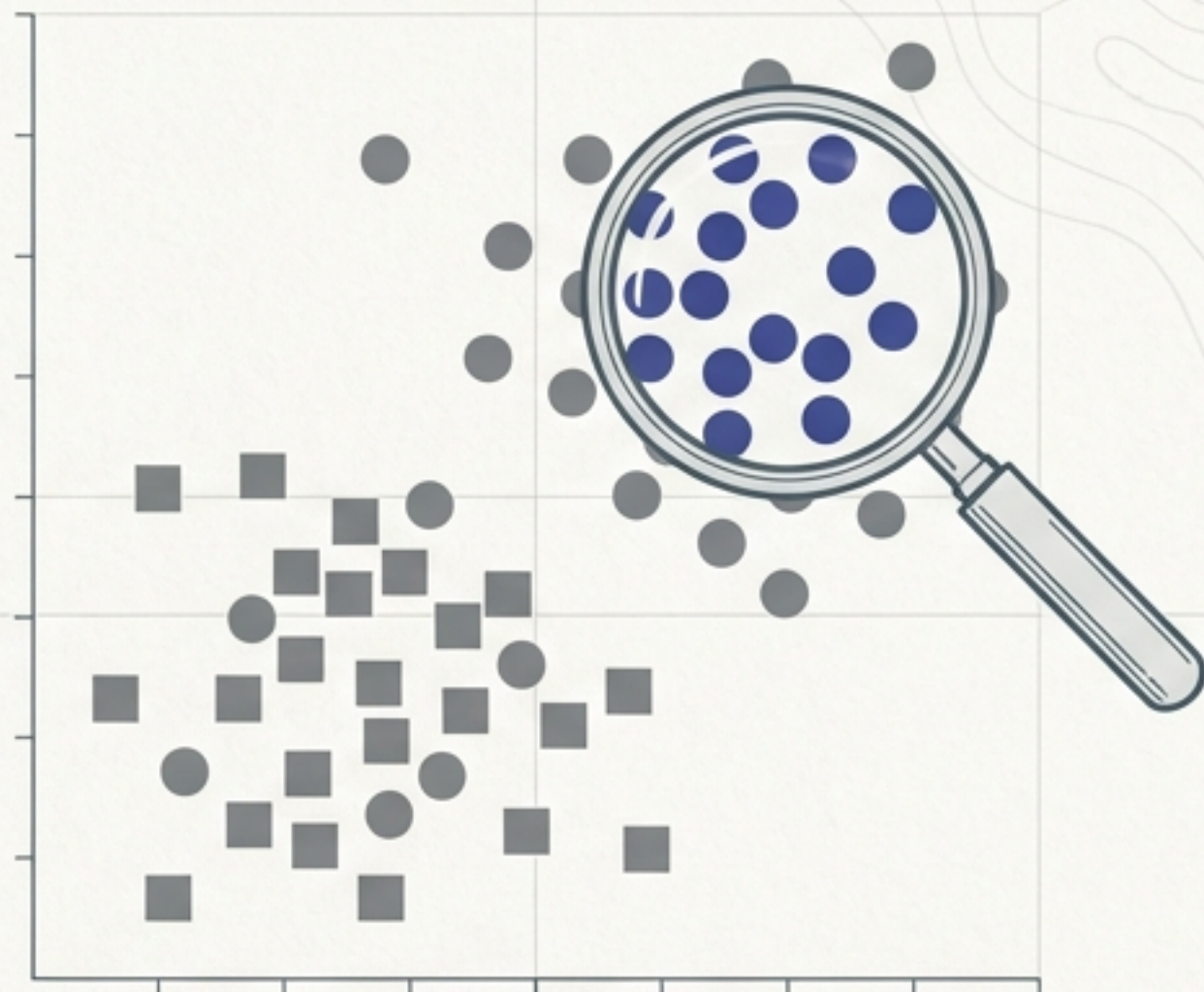
# SUPERVISED LEARNING



**Data + Labels = Prediction**

Uses a known response variable to predict future outcomes.

# UNSUPERVISED LEARNING



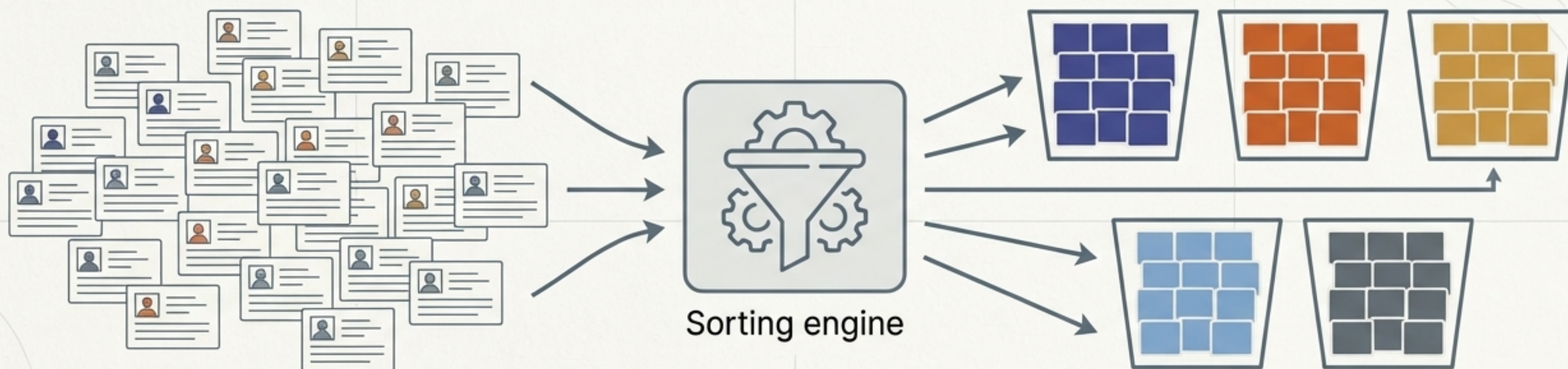
**Data Only = Discovery**

Exploratory analysis with no true answer key.  
Designed entirely to discover hidden subgroups in unmapped territory.



# The Goal of Clustering

Transforming chaos into strategic segments.



## The Business Case

A CRM Director has thousands of customer profiles but no predefined labels. The goal is to partition customers into five strategic groups for targeted management.

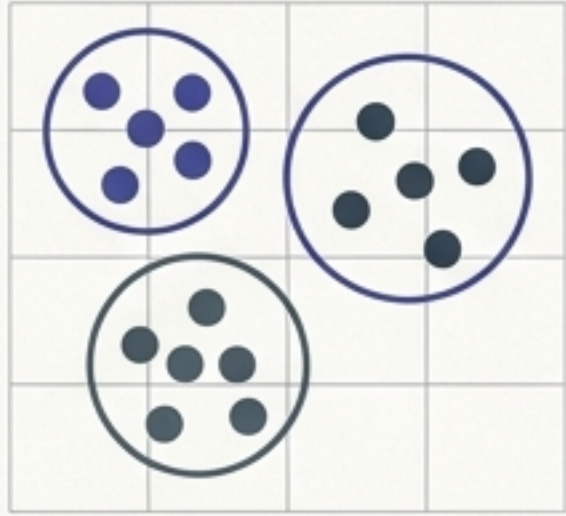
## The Golden Rule of Data Segmentation

Maximize similarities within groups.  
Maximize dissimilarities between groups.

Other Applications: Business Intelligence | Image Pattern Recognition | Web Search Topic Modeling

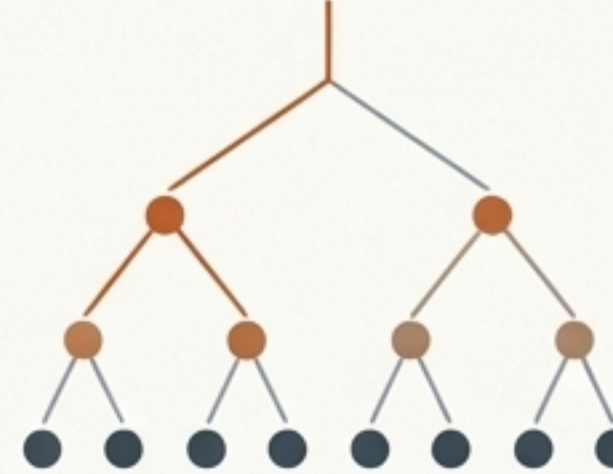


# The Four Pillars of Clustering



## Partition Methods

Assigns observations to a fixed number of unique clusters (e.g., K-Means).



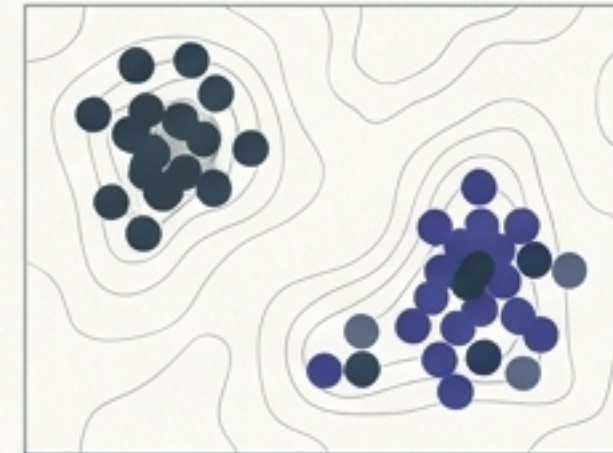
## Hierarchical Methods

Constructs a nested, tree-based representation of data similarities.



## Distribution Models

Groups data based on probability that points belong to the same distribution.



## Density Models

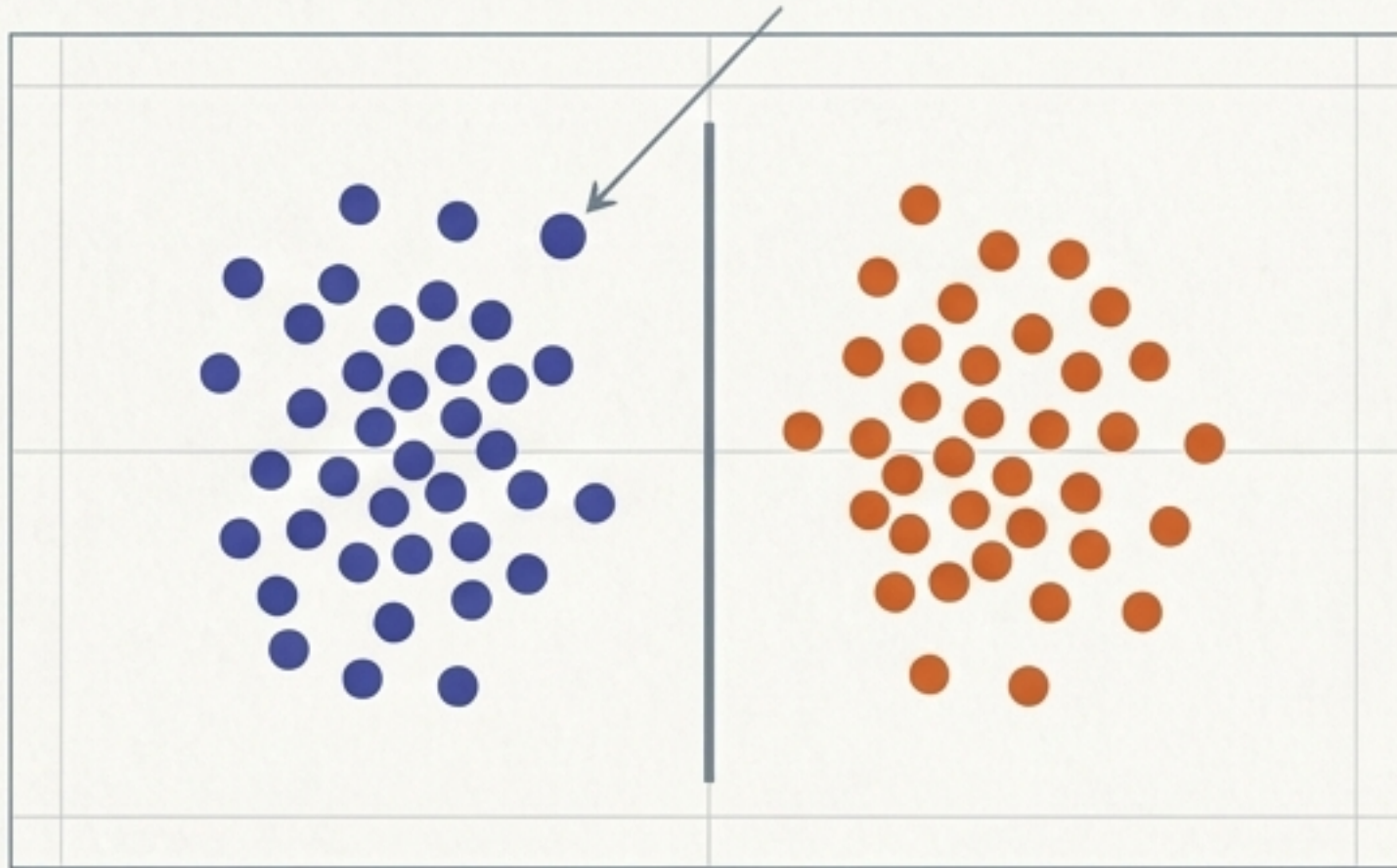
Searches for continuous areas of high-density data points (e.g., DBSCAN).

This presentation focuses entirely on Partition and Hierarchical methodologies.



# Diagnostic: Defining the Boundaries

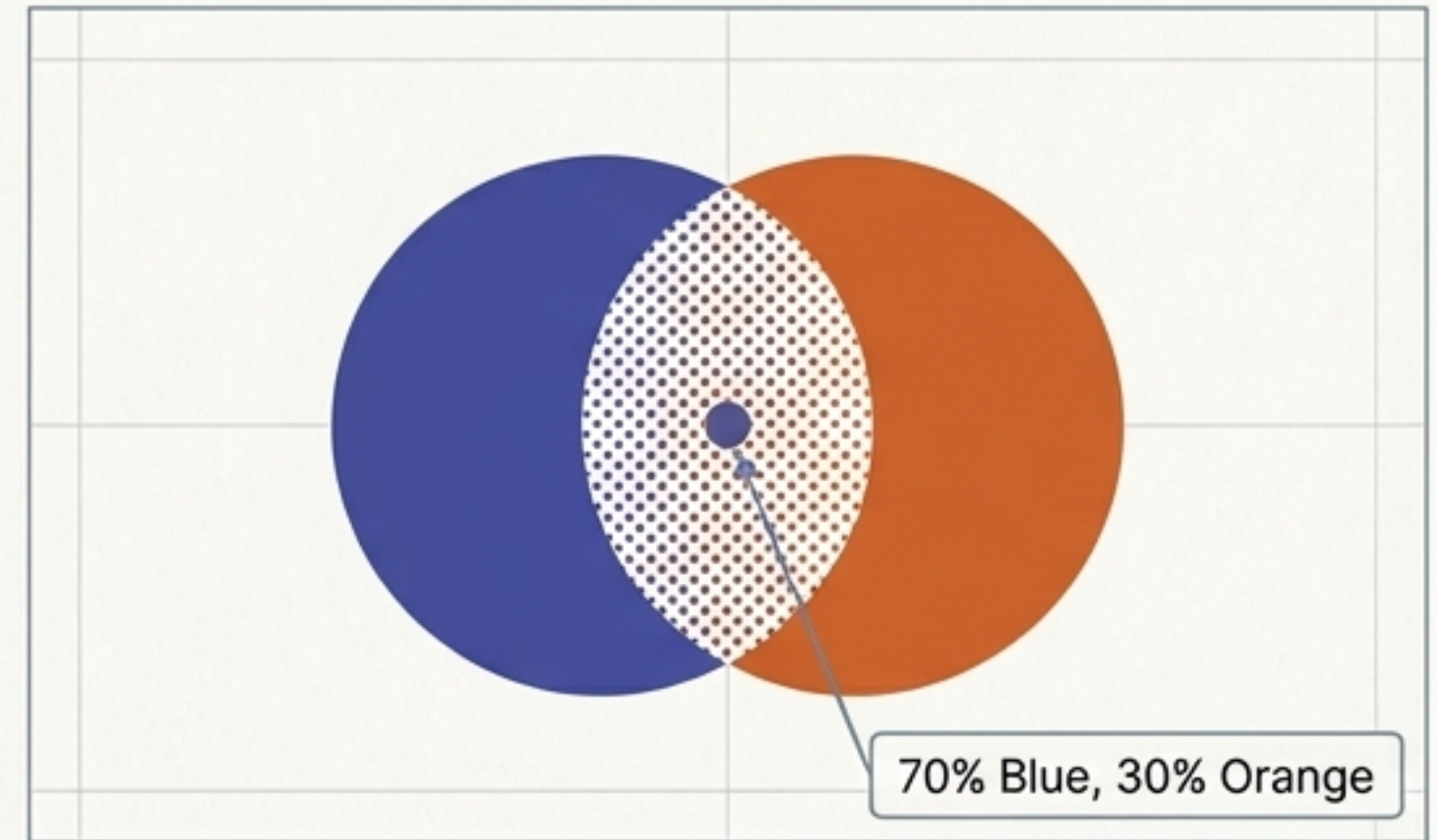
## Hard Clustering



Based on classical set theory. Subsets are mutually exclusive. An object either belongs entirely to a cluster or it does not.

**Focus: K-Means**

## Fuzzy Clustering



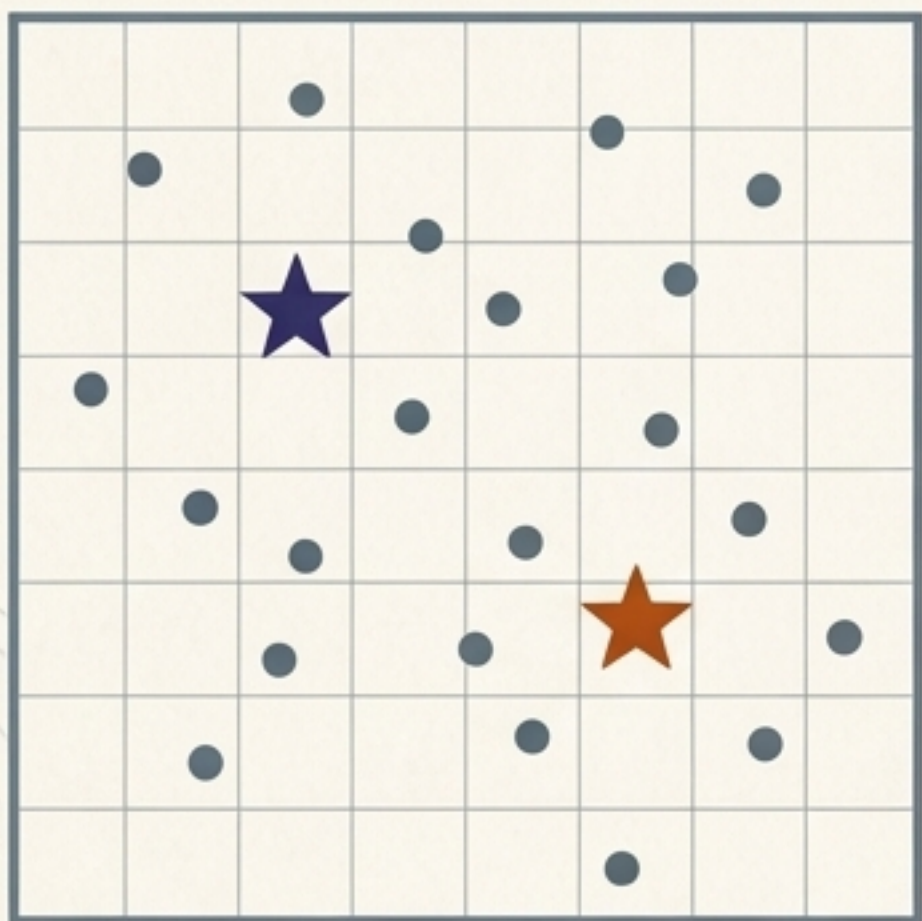
Objects belong to several clusters simultaneously with varying probabilistic degrees of membership.

**Example: Fuzzy C-Means**

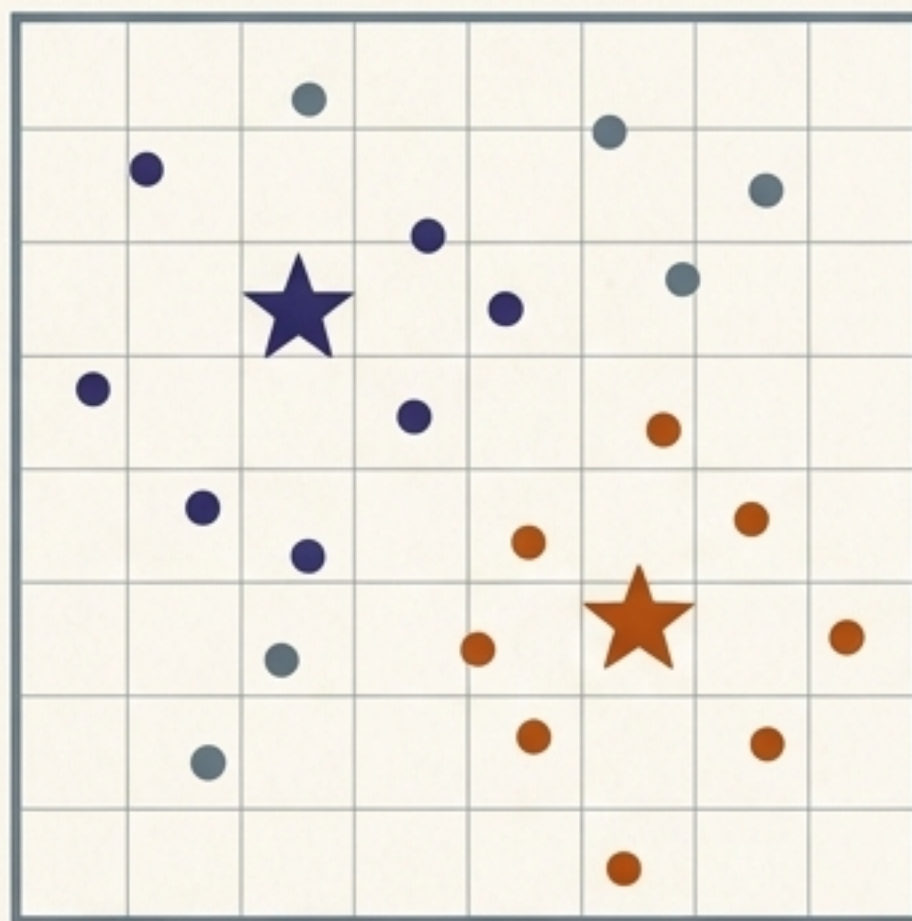
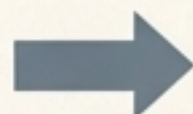


# Introduction to K-Means: The Tour Guide Metaphor

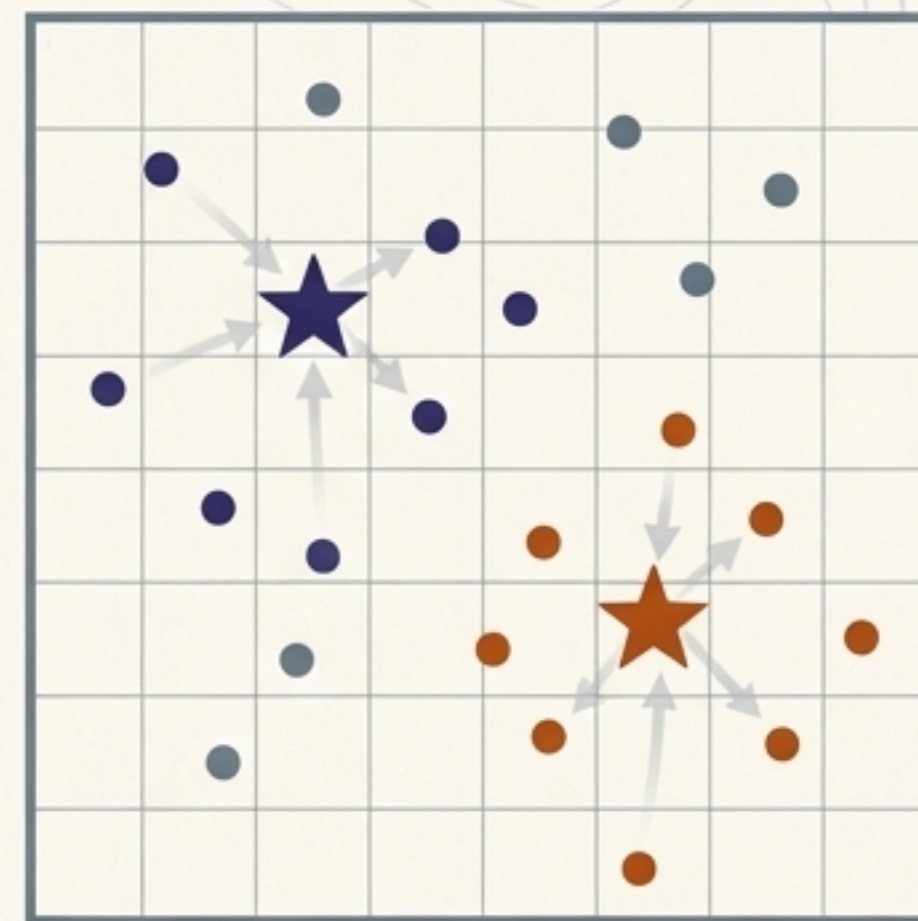
Finding the center when the destinations are unknown.



1. Tour guides (centroids) stand at random locations among the scattered tourists.



2. Tourists group themselves with the closest visible tour guide.



3. Tour guides walk to the exact center of their respective groups. The process repeats.

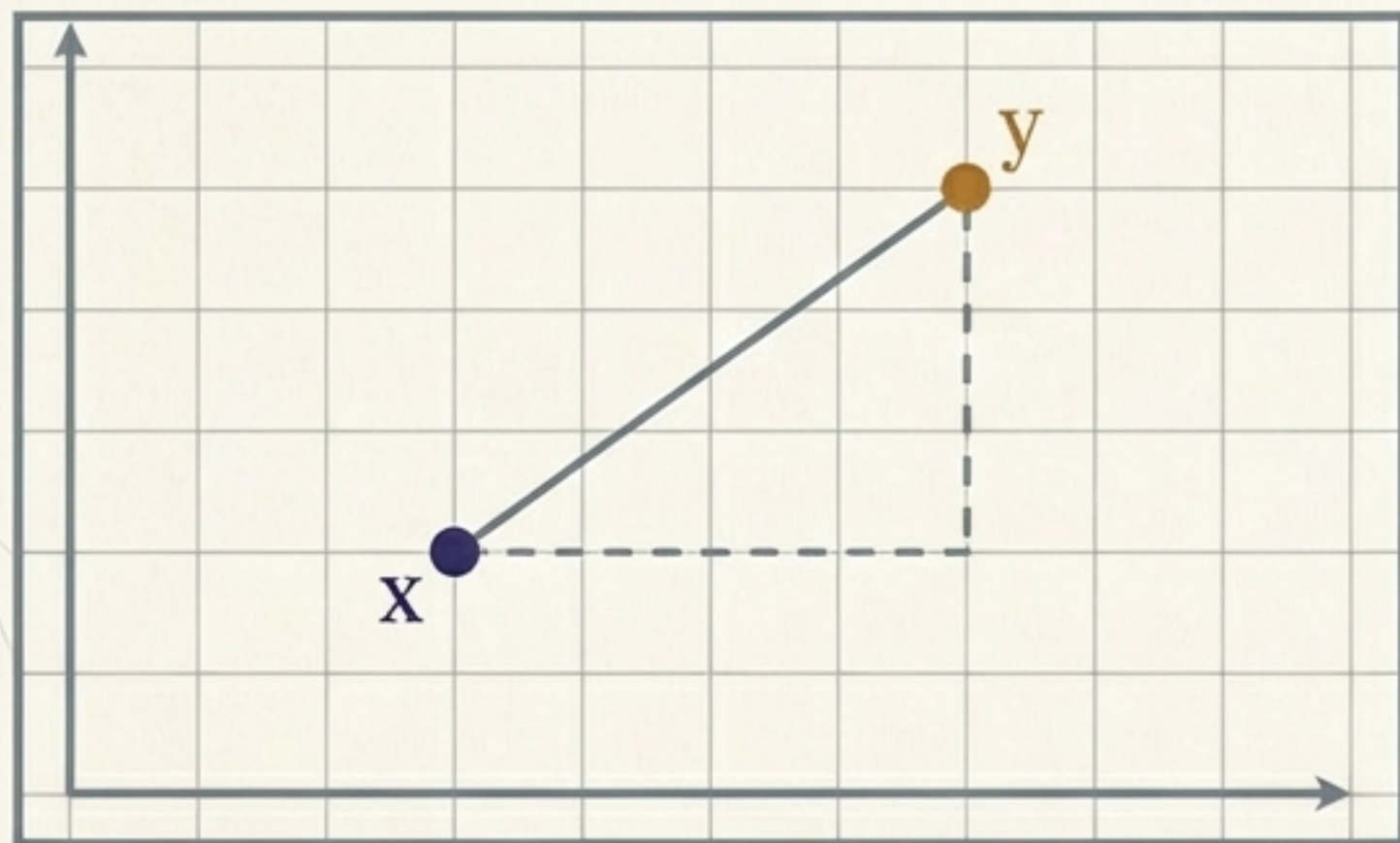
The goal of K-Means is to divide data into a known  $K$  number of categories. Because the data does not move, we must iteratively move  $K$  cluster centers until they stand exactly in the middle of their respective clusters.



# The Math Behind the Means

Two geometric prerequisites drive the algorithm.

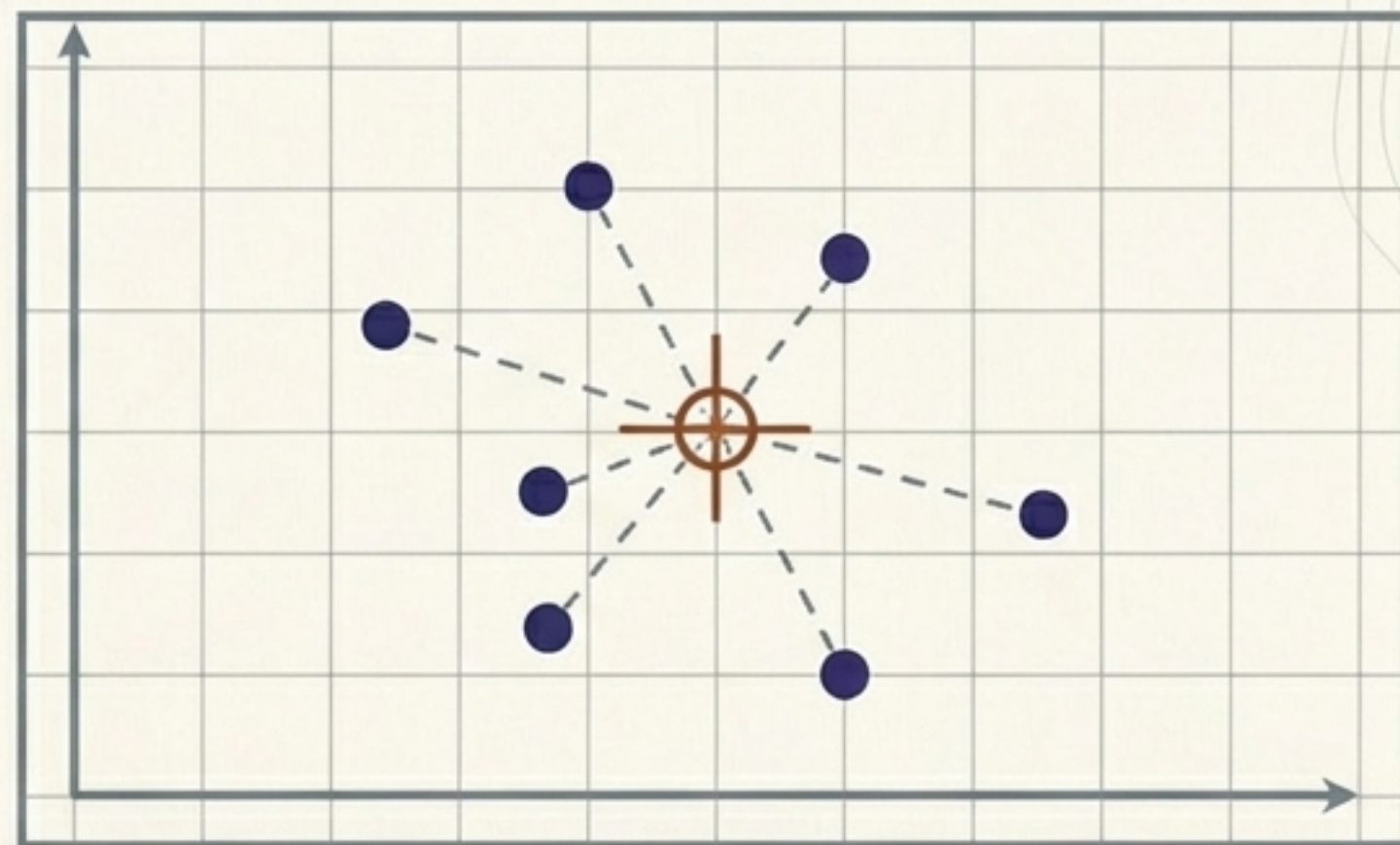
## 1. The Distance Measure



Usually Euclidean distance (sum-of-squares error) to measure spatial separation between points.

$$d^2 = (x_1 - y_1)^2 + (x_2 - y_2)^2$$

## 2. The Mean Average

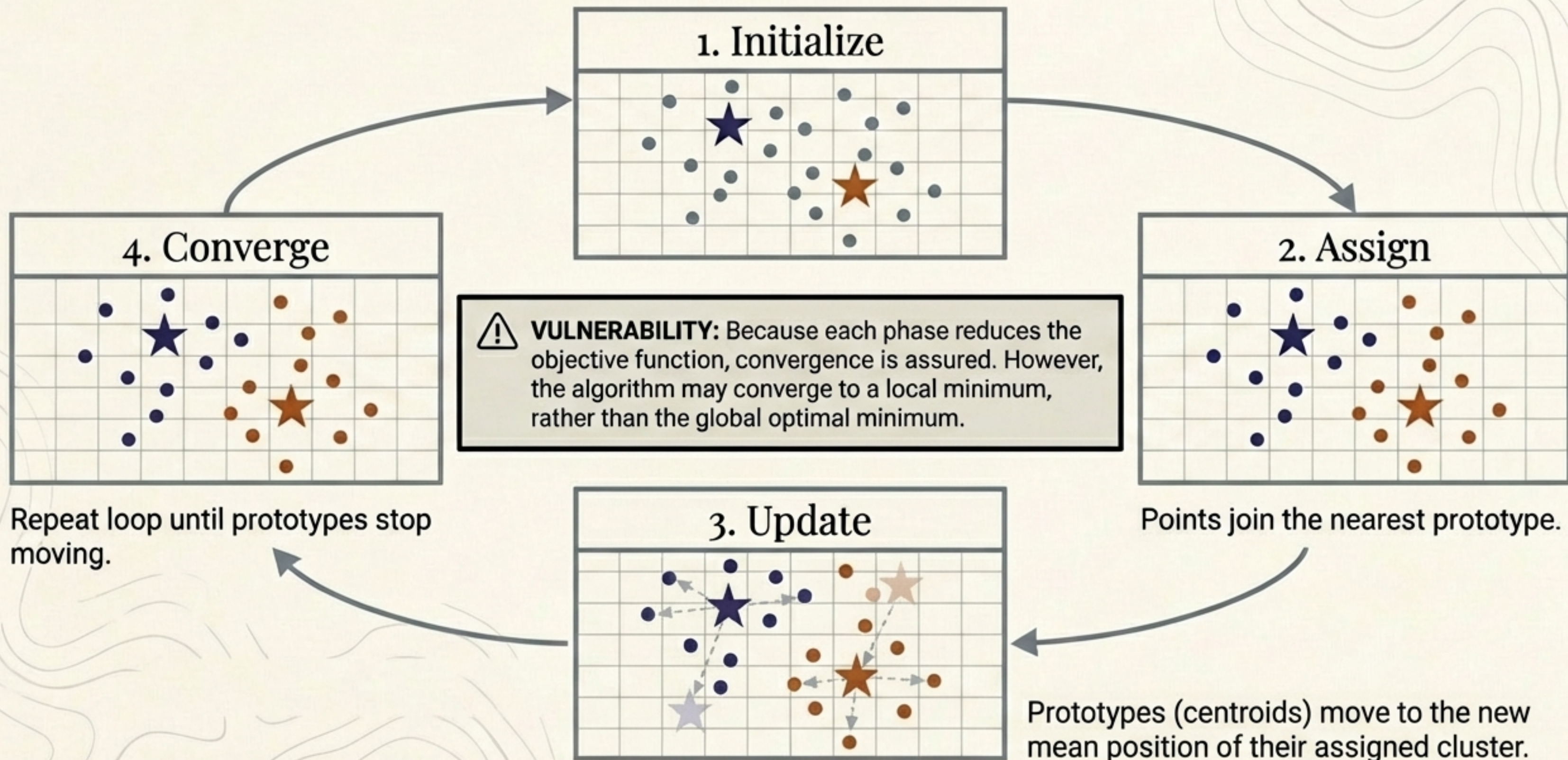


Computing the central point of assigned data in a flat space.

**Objective (J):** Minimize the distortion measure—the sum of the squares of distances from each point to its assigned vector.



# The K-Means Algorithm Cycle



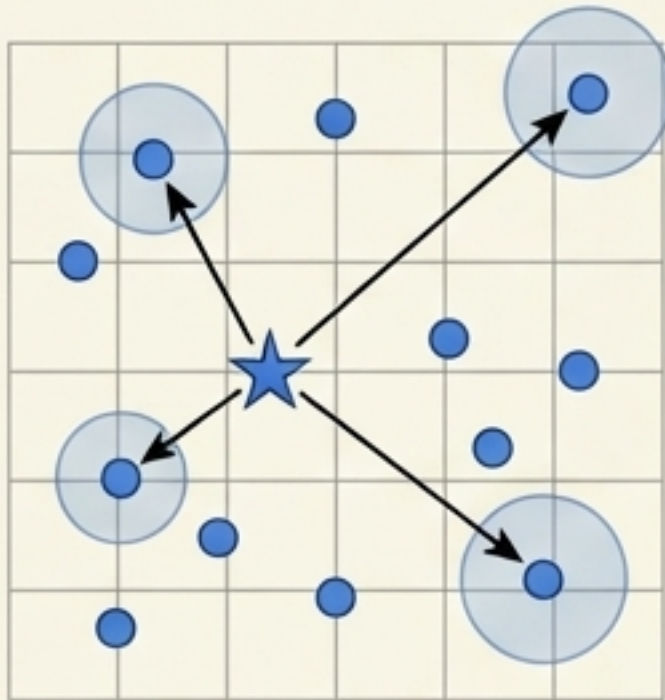


# Derivative Methods: Improving K-Means

Addressing limitations like initialization sensitivity and scalability

## K-Means++: Smarter Initialization

Selects initial centroids probabilistically, favoring points far from existing centers.  
Improves convergence speed and solution quality.

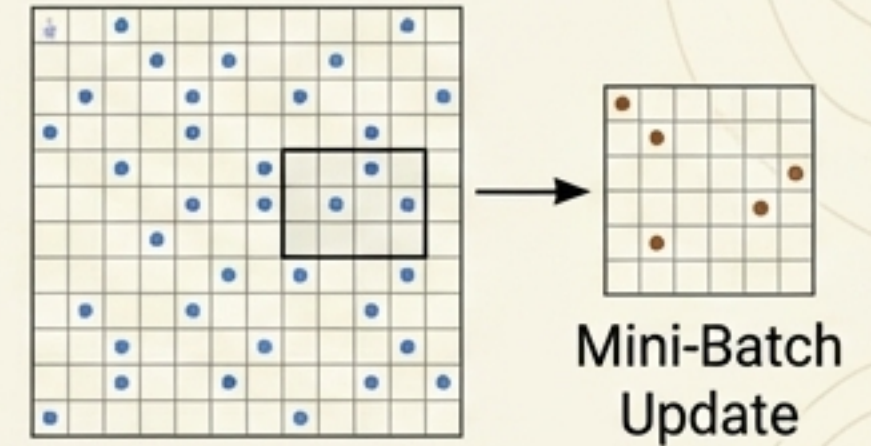


Probabilistic Selection

## Other K-Means Variants

### Mini-Batch K-Means

Uses small, random subsets of data per iteration. Faster for large datasets.



### K-Medoids (PAM)

Uses actual data points as centers (medoids).  
More robust to outliers.



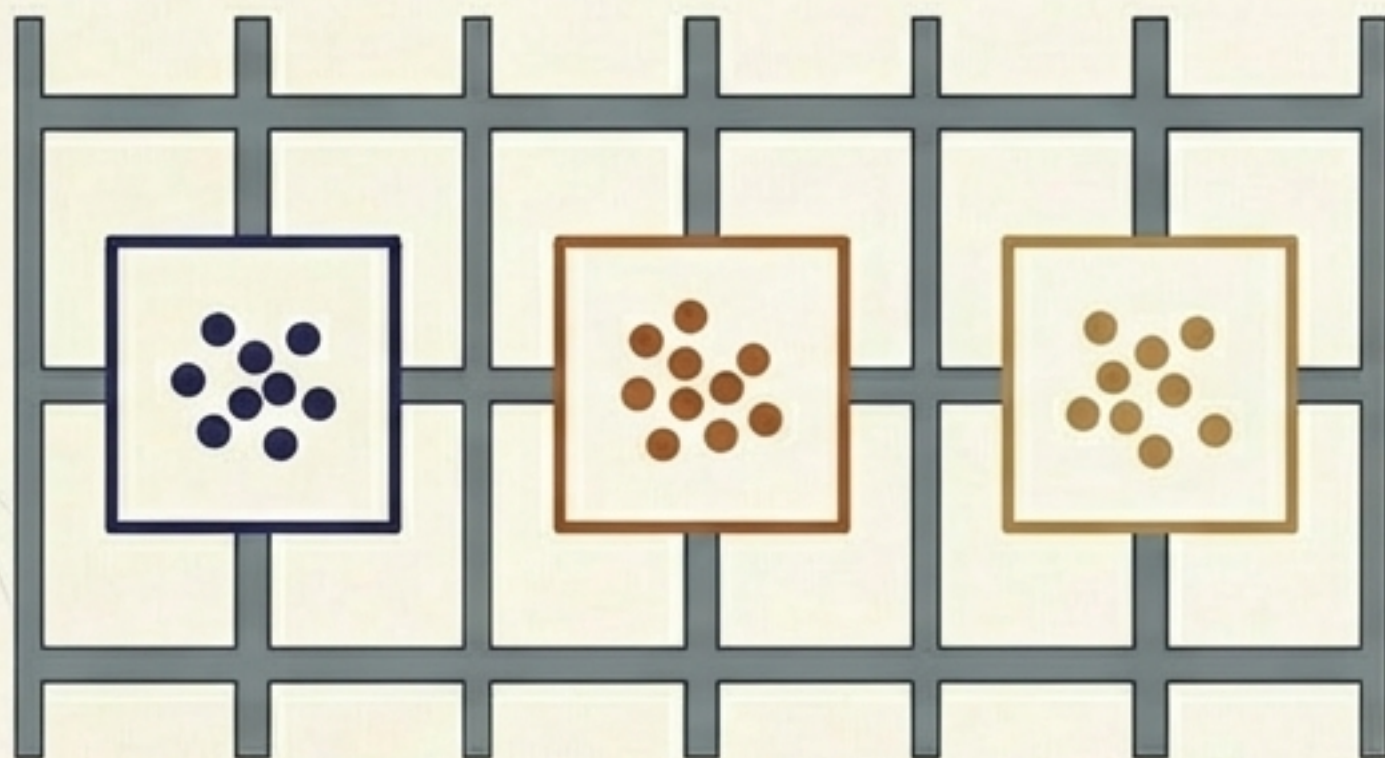
Medoid Center

Note: These methods offer trade-offs between speed, accuracy, and robustness.



# Hierarchical Clustering: An Alternative Approach

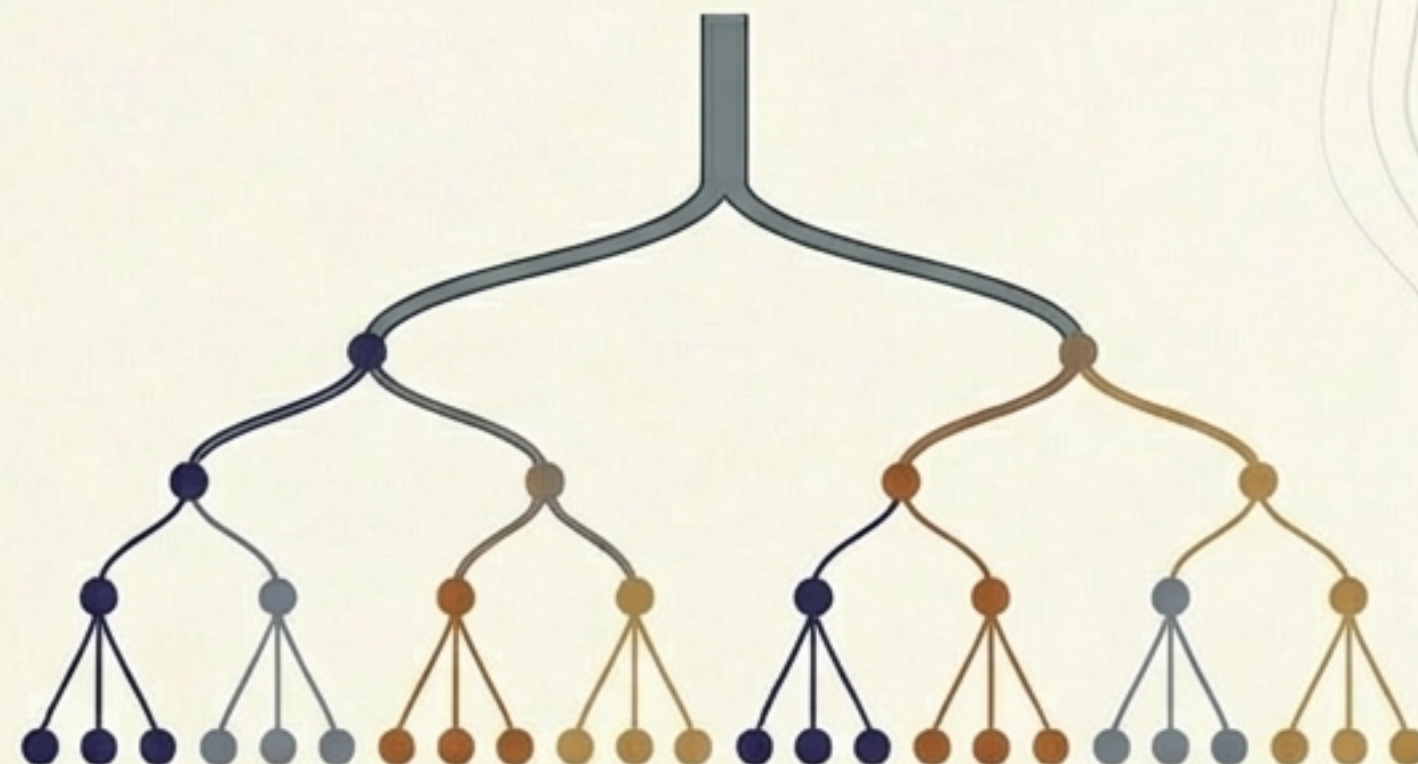
## K-Means



- Requires choosing "K" upfront.
- Forces data into pre-set buckets.

VS

## Hierarchical (Agglomerative)

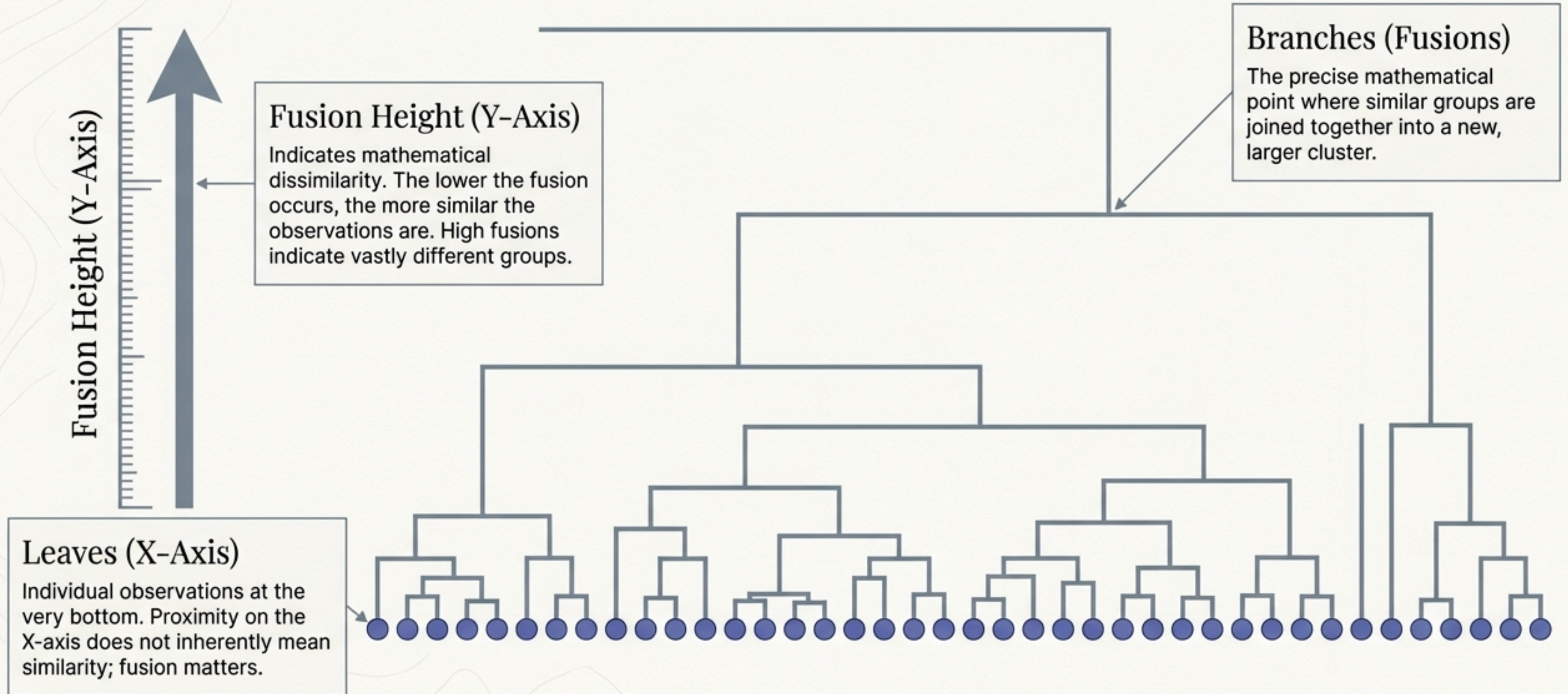


- Exploratory: No "K" needed.
- Builds a nested, tree-based representation.

Instead of committing to a pre-defined number of partitions, agglomerative (bottom-up) clustering reveals the natural, underlying taxonomy of the dataset.

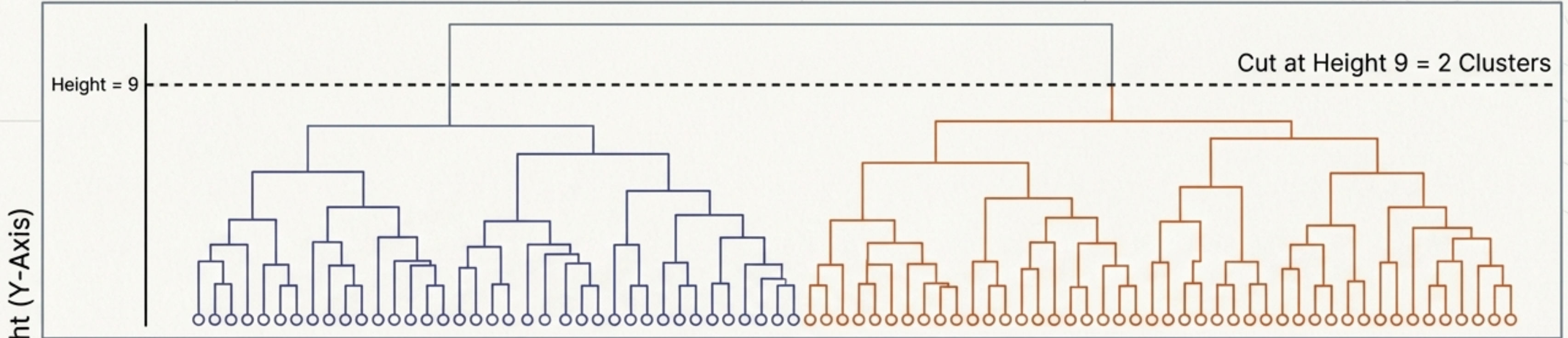


# Anatomy of a Dendrogram





# Pruning the Tree: Extracting Clusters



**Hierarchical Nestedness:** The horizontal cut dictates the final  $K$ . Clusters obtained at lower heights are permanently nested within the larger clusters obtained at greater heights.



# The Agglomerative Algorithm

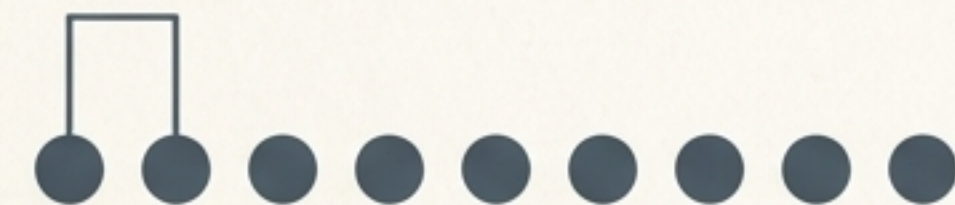
A bottom-up process of iterative fusion.

Start:  $n$  clusters



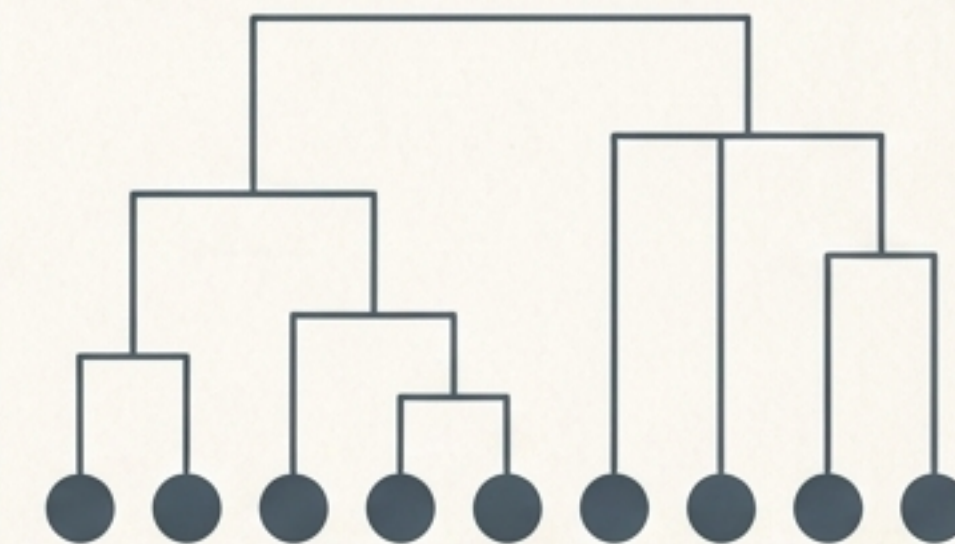
Treat every single observation as its own independent cluster.

Step 1:  $n-1$  clusters



Calculate pairwise dissimilarities. Fuse the least dissimilar (closest) pair.

Repeat until 1 trunk

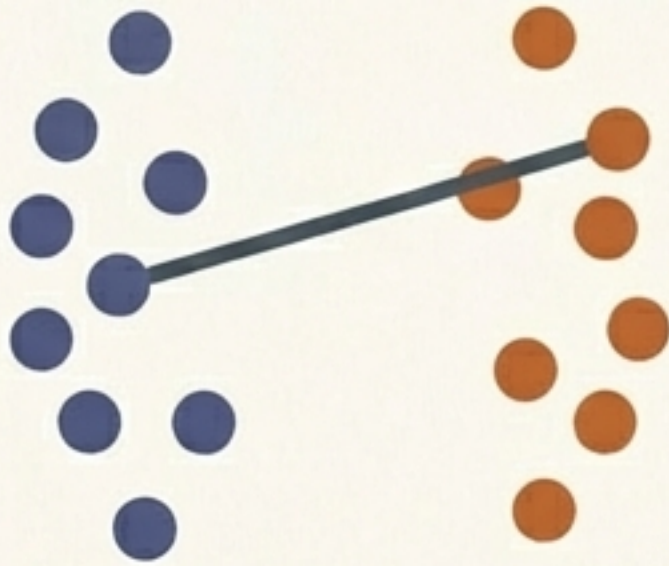


Continuously fuse the next closest groups until only one master cluster remains.



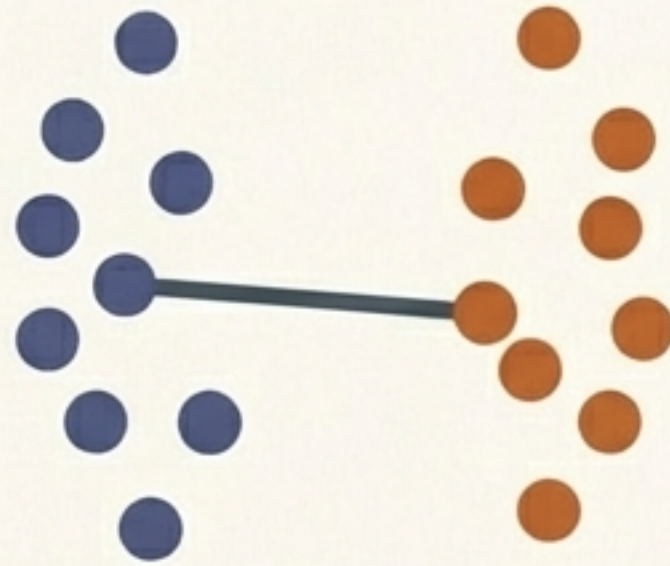
# Measuring Cluster Distance: Linkage Methods

Complete  
(Maximal)



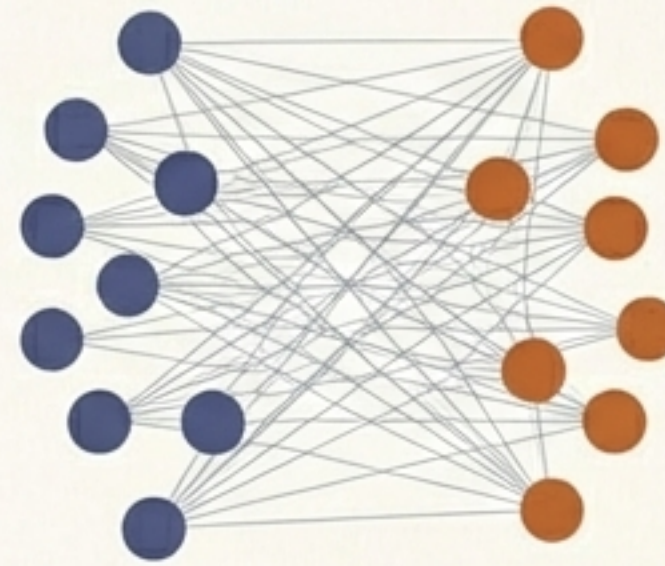
Measures the largest dissimilarity. Preferred for balanced trees.

Single  
(Minimal)



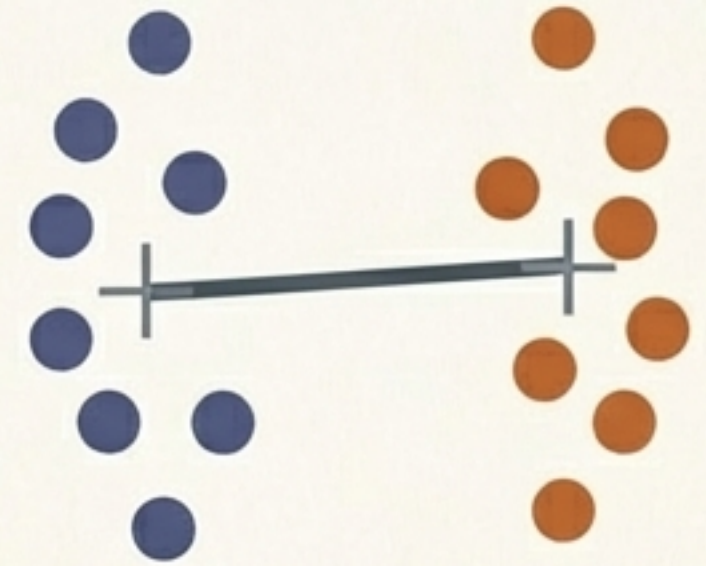
Measures the smallest dissimilarity. Can result in trailing, extended clusters.

Average  
(Mean)



Measures the mean of all pairwise dissimilarities. Preferred for balanced trees.

Centroid



Measures dissimilarity between the geometric centroids. Often used in genomics.



# Synthesis: Choosing the Right Tool

| Dimension              | K-Means Partitioning                           | Hierarchical Agglomeration                      |
|------------------------|------------------------------------------------|-------------------------------------------------|
| Number of Clusters (K) | Pre-specified upfront.                         | Exploratory. Determined post-hoc via tree cuts. |
| Output Shape           | Flat, mutually exclusive spatial partitions.   | Nested, highly visual tree-based taxonomy.      |
| Computational Load     | Fast and highly scalable for massive datasets. | Computationally heavy. Slower for large data.   |

**FINAL TAKEAWAY:** Use K-means when you already know your target segments. Use Hierarchical when you need to map the undiscovered taxonomy of your data.